

Use Case Specifications

Online Internship & Placement Prep Portal

Basic flow and alternate flow for each use case in the use case diagram

1. Register & Login

Primary actor(s): Student, Admin (shared)

Basic flow

1. User navigates to the portal's landing page.
2. User selects "Register" (new user) or "Login" (existing user).
3. For registration: user enters name, email, password, and role-specific details (e.g., college/batch for students).
4. System validates input format (email syntax, password strength).
5. System checks that the email is not already registered.
6. System creates the account and stores the password in hashed form.
7. System redirects the user to the login page with a success message.
8. User enters email/username and password on the login page.
9. System validates credentials against the stored hashed values.
10. System creates an authenticated session and redirects the user to their respective dashboard (Student or Admin).

Alternate flow

A1 — Invalid login credentials

At step 9, if credentials do not match, the system displays "Invalid email or password" and returns the user to the login form (step 8) without creating a session.

A2 — Duplicate registration email

At step 5, if the email is already registered, the system displays "Account already exists" and offers a link to the login page instead of creating a new account.

2. View Internship Listings

Primary actor(s): Student

Basic flow

1. Authenticated student navigates to the "Internships" section.
2. System retrieves active internship listings posted by admin from the database.
3. System displays listings with title, company, and short description.
4. Student applies filters (location, domain, stipend, duration) as needed.
5. System re-queries and refreshes the listing view based on the filters.
6. Student selects a specific listing to view its details.
7. System displays the full description, eligibility, deadline, and current application status (if already applied).
8. Student may bookmark/save a listing for later reference.
9. Student navigates back to the full listings page.
10. Student proceeds to apply directly from the detail view, triggering the Apply for Internship use case.

Alternate flow

A1 — No listings available

At step 3, if no active listings exist, the system displays "No internships currently available."

A2 — Filter returns no results

At step 5, if the filter criteria match no listings, the system displays "No matching internships found" and lets the student reset the filters.

3. Apply for Internship

Primary actor(s): Student

Includes: *View Internship Listings*

Basic flow

1. Student views internship listings (include: View Internship Listings).
2. Student selects a listing and clicks "Apply".
3. System checks whether the student has already applied to this listing.
4. System displays the application form (resume upload, cover note, etc.) if not already applied.
5. Student fills in / reviews the required application details.
6. Student submits the application.
7. System validates that all mandatory fields and attachments are present.
8. System records the application with a timestamp and an "Applied" status.
9. System updates the listing's application count for the admin view.
10. System displays a confirmation and adds the entry under "My Applications".

Alternate flow

A1 — Duplicate application blocked

At step 3, if the student has already applied, the system blocks the action and displays "You have already applied to this internship" instead of showing the form.

A2 — Incomplete application

At step 7, if mandatory fields are missing, the system highlights the missing fields and returns to step 5 without submitting.

4. Practice Questions (Aptitude / MCQ / Coding)

Primary actor(s): Student

Basic flow

1. Student navigates to the "Practice" section.
2. System displays available categories (Aptitude, Technical MCQ, Coding).
3. Student selects a category.
4. System retrieves questions from the categorized question bank matching that category.
5. System displays the first question with answer options / code editor as applicable.
6. Student submits an answer for the current question.
7. System evaluates the answer and shows immediate correctness feedback.
8. Student proceeds to the next question (steps 6–7 repeat until the set is exhausted or the student stops).
9. System compiles the practice session score once the student completes or ends the set.
10. System stores the session results and updates the student's practice history for analytics.

Alternate flow

A1 — No questions in category

At step 4, if the question bank has no questions for the selected category, the system displays "No questions available in this category yet" and returns to step 2.

A2 — Student exits mid-session

At step 8, if the student exits before completing the set, the system saves partial progress and results up to the last answered question rather than discarding the session.

5. Take Timed Mock Test

Primary actor(s): Student

Includes: *Auto-Submit on Timeout, View Score & Answer Analysis*

Basic flow

1. Student navigates to the "Mock Tests" section.
2. Student selects a mock test (by category, difficulty, or duration).
3. System displays the test instructions and time limit.
4. Student clicks "Start Test" and the system starts the countdown timer.
5. System presents questions with navigation controls to move between them.
6. Student answers questions, marking/reviewing/changing answers within the time limit.
7. Student clicks "Submit Test" before time expires.
8. System locks further input and records all submitted answers.
9. System calculates the score and correct/incorrect breakdown (include: View Score & Answer Analysis).
10. System stores the test attempt in the student's history for analytics.

Alternate flow

A1 — Time expires before manual submission

At step 7, if the timer reaches zero, the system automatically submits the test (include: Auto-Submit on Timeout) using whatever answers were recorded, then proceeds to step 8.

A2 — Connectivity loss during the test

At step 6, if the student's connection drops, the system preserves already-recorded answers and resumes the same session with remaining time when the student reconnects, rather than restarting the test.

6. Auto-Submit on Timeout

Primary actor(s): System (triggered on behalf of Student)

Basic flow

1. System continuously tracks elapsed time against the test's configured duration.
2. Timer reaches zero while the student is still attempting the test.
3. System immediately disables all answer input controls.
4. System captures the current state of all answered and unanswered questions.
5. System marks the test attempt status as "Auto-Submitted".
6. System records the submission timestamp as the configured end time.
7. System processes the recorded answers the same way as a manual submission.
8. System notifies the student on-screen that the test was auto-submitted due to timeout.
9. System proceeds to scoring and analysis generation.
10. System logs the auto-submit event for audit and analytics purposes.

Alternate flow

A1 — Student had just manually submitted

If the system detects the attempt already has a "Submitted" status at the moment the timer expires (race condition), it skips the auto-submit process to avoid creating a duplicate record.

7. View Score & Answer Analysis

Primary actor(s): Student

Basic flow

1. Student completes/submits a mock test, or navigates from "My Test History".
2. System retrieves the recorded answers for the selected attempt.
3. System compares each recorded answer against the correct answer key.
4. System computes the total score, accuracy percentage, and time taken.
5. System displays a results summary (score, rank/percentile if applicable).
6. Student clicks into "Detailed Analysis".
7. System displays a question-by-question breakdown (correct / incorrect / skipped) with explanations.
8. Student filters the breakdown by topic or difficulty.
9. Student reviews individual questions and their correct solutions.
10. Student navigates back to test history or proceeds to the analytics dashboard.

Alternate flow

A1 — Analysis not yet generated

At step 3, if scoring is still being computed (e.g., delayed evaluation for coding questions), the system displays "Results are being processed" and lets the student check back later.

8. View Personal Analytics Dashboard

Primary actor(s): Student

Includes: *View Score & Answer Analysis*

Basic flow

1. Student navigates to the "Analytics" / "Dashboard" section.
2. System retrieves the student's full test and practice history.
3. System aggregates scores across attempts to compute performance trends over time.
4. System identifies strength and weakness areas by topic/category.
5. System renders trend charts (e.g., score over time, accuracy by category).
6. Student drills down into a specific test attempt for full details (include: View Score & Answer Analysis).
7. Student filters the dashboard by date range or test type.
8. System recalculates and refreshes the visualizations based on the filter.
9. Student views recommended focus areas generated from the weakness analysis.
10. Dashboard auto-updates the next time the student completes a new test or practice session.

Alternate flow

A1 — Insufficient history

At step 3, if the student has no completed tests or practice sessions yet, the system displays "Complete a test or practice session to see your analytics" instead of charts.

9. Manage Internship Listings

Primary actor(s): Admin

Basic flow

1. Admin navigates to "Manage Internships".
2. System displays existing listings with status (active / expired / draft).
3. Admin selects "Add New Listing".

4. Admin enters listing details (title, company, description, eligibility, deadline, stipend).
5. Admin submits the listing.
6. System validates required fields and deadline validity (must be a future date).
7. System saves the listing and marks it "Active", making it visible to students.
8. Admin selects an existing listing to edit its details.
9. Admin saves changes; system updates the listing and timestamps the modification.
10. Admin deactivates/removes a listing; system hides it from student view while retaining historical application records.

Alternate flow

A1 — Validation failure

At step 6, if required fields are missing or the deadline is in the past, the system rejects the submission, highlights the errors, and returns to step 4.

A2 — Removing a listing with active applications

At step 10, if the listing has pending student applications, the system warns the admin and asks for confirmation before deactivating it.

10. Manage Categorized Question Bank

Primary actor(s): Admin

Basic flow

1. Admin navigates to the "Question Bank" section.
2. System displays existing categories (Aptitude, Technical MCQ, Coding) with question counts.
3. Admin selects a category or creates a new one.
4. Admin selects "Add Question".
5. Admin enters question content, options/test cases, correct answer, and difficulty level.
6. Admin submits the question.
7. System validates that all required fields (e.g., correct answer, options for MCQ) are present.
8. System saves the question to the selected category, making it available for practice and tests.
9. Admin edits or deletes existing questions from the bank.
10. System updates the question bank and reflects the change immediately.

Alternate flow

A1 — Incomplete question data

At step 7, if a required field (e.g., no correct answer marked) is missing, the system rejects the submission and returns to step 5 with the error highlighted.

A2 — Deleting a question in use

At step 9, if the question is currently part of an active or scheduled mock test, the system warns the admin before allowing deletion.

11. View Student Applications

Primary actor(s): Admin

Basic flow

1. Admin navigates to the "Applications" section.
2. System retrieves all applications submitted across all internship listings.
3. System displays applications in a table with student name, listing, and status.
4. Admin filters applications by listing, date, or status.

5. System refreshes the table based on the filter.
6. Admin selects a specific application to view full details.
7. System displays the student's submitted resume/cover note and profile information.
8. Admin updates the application status (e.g., Shortlisted, Rejected, Selected).
9. System saves the status change and timestamps it.
10. System reflects the updated status to the student under "My Applications".

Alternate flow

A1 — No applications yet

At step 3, if no student has applied to any listing, the system displays "No applications received yet."

A2 — Bulk status update

At step 8, the admin selects multiple applications and applies a status change in bulk; the system validates and updates all selected records in a single operation.